

D488

Cybersecurity Architecture and Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional credit union spent \$400,000 last year on a next-generation firewall that its vendor rates in the top tier. An employee's laptop was infected through a malicious email attachment and began probing internal servers. Which design principle, if applied, would most likely have limited the spread?

- A. Layered controls behind the firewall, such as endpoint protection and internal network segmentation.
- B. Encrypting the internal servers' disks.
- C. Replacing the firewall with an even higher-rated model.
- D. Relying on the perimeter firewall, because it blocks all malicious traffic.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A manufacturing company with plants in three states lets any device connected to its headquarters network reach the payroll system without another sign-in. The network team recently upgraded to faster switches. Under a zero trust architecture, what change is required?

- A. Authenticate and authorize every request to payroll using user identity, device health, and least privilege, wherever the request comes from.
- B. Enable port security on the new switches.
- C. Verify users once at morning login and trust the session for the rest of the day.
- D. Allow payroll access only from the headquarters network and block remote users.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

Which situation violates zero trust principles?

- A. An engineer must pass MFA again before opening a production database from the office.
- B. A contractor on the office VPN gets access to all internal file shares because the VPN already authenticated the device.
- C. An app's session is cut off when the laptop fails a device-health check.
- D. A user receives only the permissions needed for her current project.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A hospital stores patient records on servers with full-disk encryption that passed last year's audit with no findings. Nurses view the records on tablets over the hospital's wireless network, which covers all six floors. Which control protects the records while they travel to the tablets?

- A. Stronger passwords for the server administrators.
- B. Nothing more, because the audit found no issues.
- C. The full-disk encryption already on the servers.
- D. Encrypting the connection with TLS.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A team encrypts a database's storage volumes with keys held in a key management service. Which data state does this control mainly protect?

- A. Data in transit.
- B. Data in use.
- C. Data at rest.
- D. All three states, because encryption follows the data wherever it goes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A retail startup with 40 employees moves its order system to an infrastructure-as-a-service (IaaS) provider whose data centers hold several major certifications. A month later, an unpatched operating system on one of its virtual machines is flagged. Who is responsible for patching that operating system?

- A. The provider, because its certifications cover the customer's systems.
- B. The startup, because in IaaS the customer manages the guest operating system.
- C. The provider, because it owns the physical servers.
- D. Both equally, split by default in every contract.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

In which cloud service model does the customer's security responsibility shrink mainly to its data, user accounts, and access settings?

- A.** Platform as a service (PaaS).
- B.** Infrastructure as a service (IaaS).
- C.** None; in every model the provider also secures the customer's user accounts.
- D.** Software as a service (SaaS).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

During an assessment, an analyst at a mid-size insurer notes that a public web server runs software with a known, unpatched flaw. The server sits in a data center that was renovated in 2024. In risk terms, what is the unpatched flaw?

- A.** A vulnerability.
- B.** A risk.
- C.** A threat.
- D.** An incident, because a known flaw means the server is already compromised.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A security team at a logistics firm confirms that ransomware is encrypting files on two file servers. The firm's cyber insurance renews next month, and the CEO is traveling. Following the standard incident response lifecycle, what should the team do next?

- A.** Write the lessons-learned report.
- B.** Contain it by isolating the affected servers from the network.
- C.** Restore the servers from backups right away.
- D.** Confirm insurance coverage with the CEO before taking any action.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

An incident team has isolated an infected host and removed the malware and the account it used. Which phase comes next?

- A.** Post-incident lessons learned.
- B.** Preparation.
- C.** Recovery: restore the system to service and monitor it closely.
- D.** Detection and analysis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A cloud hosting company is choosing a hypervisor for 300 production servers. Its lead architect prefers the vendor with the best-rated support desk. Security wants the smallest attack surface and no general-purpose host operating system under the virtual machines. Which choice fits?

- A.** Whichever product has the best-rated support desk.
- B.** Containers on a shared host kernel, which always isolate workloads more strongly than virtual machines.
- C.** A Type 2 (hosted) hypervisor.
- D.** A Type 1 (bare-metal) hypervisor.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A software team at a payments company runs a security scan only in the week before each release. Last quarter it shipped 14 releases, and late findings delayed three of them. Which change best lowers the cost of fixing security defects?

- A.** Drop the scan so releases are no longer delayed.
- B.** Hire an outside firm for a larger penetration test before each release.
- C.** Add threat modeling during design and automated security tests on every code commit.
- D.** Ship fewer releases so each pre-release scan has more time.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A regional credit union ~~spent \$400,000 last year on a next-generation firewall that its vendor rates in the top tier.~~ An employee's laptop was infected through a malicious email attachment and began probing internal servers. Which design principle, if applied, would most likely have limited the spread?

Target word: principle that limits spread after one control is bypassed **Decides:** The threat arrived by email and was already inside the perimeter.

A	Correct	Defense in depth puts independent controls inside the perimeter, so one bypass does not open every server.
B	Adjacent category	Disk encryption protects stored data if media is stolen, not a live infected host moving across the network.
C	Camouflage	Answers the spend and vendor rating; a perimeter device does not inspect traffic that starts inside.
D	Absolute language	No single control blocks everything; the infection already got past it.

2. Answer A Concept 2

A manufacturing company ~~with plants in three states~~ lets any device connected to its headquarters network reach the payroll system without another sign-in. ~~The network team recently upgraded to faster switches.~~ Under a zero trust architecture, what change is required?

Target word: zero trust change **Decides:** Access is currently granted by network location alone.

A	Correct	Zero trust removes location-based trust and checks each request.
B	Camouflage	Answers the switch upgrade; port security does not verify who is accessing payroll.
C	Dropped element	Drops continuous re-evaluation of each access.
D	Adjacent category	That is still perimeter trust based on location, the model zero trust replaces.

3. Answer B Concept 2 · transfer

Which situation violates zero trust principles?

Target word: find the violation — reverse of applying zero trust **Decides:** Access granted because a connection is already on the VPN.

A	Inverted operation	That is zero trust working as intended.
B	Correct	Trust is inherited from network location, and access is far broader than needed.
C	Confusion set	Continuous device checks are part of zero trust, not a violation.
D	Adjacent category	Least privilege supports zero trust.

4. Answer D Concept 3

A hospital stores patient records on servers with full-disk encryption ~~that passed last year's audit with no findings~~. Nurses view the records on tablets over the hospital's wireless network, ~~which covers all six floors~~. Which control protects the records while they travel to the tablets?

Target word: control for data in transit **Decides:** The exposure is while records travel over the network.

A	Adjacent category	An access control on the servers, not protection of data in motion.
B	Camouflage	A clean audit of stored data says nothing about data on the wireless network.
C	Confusion set	Disk encryption protects data at rest; data leaving the server is decrypted for transport.
D	Correct	Channel encryption protects data in transit.

5. Answer C Concept 3 • transfer

A team encrypts a database's storage volumes with keys held in a key management service. Which data state does this control mainly protect?

Target word: name the state from the control — reverse direction **Decides:** Storage-volume encryption protects stored data.

A	Confusion set	Moving data needs TLS or IPsec, not volume encryption.
B	Adjacent category	Data being processed is decrypted in memory; it needs access and memory protections.
C	Correct	Volume encryption protects data stored on disk.
D	Absolute language	Each state needs its own control.

6. Answer B Concept 4

A retail startup ~~with 40 employees~~ moves its order system to an infrastructure-as-a-service (IaaS) provider ~~whose data centers hold several major certifications~~. A month later, an unpatched operating system on one of its virtual machines is flagged. Who is responsible for patching that operating system?

Target word: owner of guest OS patching **Decides:** In IaaS the customer manages the guest operating system.

A	Camouflage	Provider certifications cover the provider's controls, not the customer's configuration.
B	Correct	The provider secures hardware and the virtualization layer; the customer owns the OS and up.
C	Adjacent category	Owning hardware covers the physical and hypervisor layers, not the guest OS.
D	Confusion set	Responsibility is divided by layer, not split evenly.

7. Answer D Concept 4 • transfer

In which cloud service model does the customer's security responsibility shrink mainly to its data, user accounts, and access settings?

Target word: service model from the responsibility — reverse **Decides:** The provider runs the whole application stack.

A	Adjacent category	In PaaS the customer still owns its application code and configuration.
B	Inverted operation	IaaS gives the customer the most responsibility, from OS upward.
C	Absolute language	Identity and access management always stays with the customer.
D	Correct	The provider runs the application; the customer keeps data, identities, and access.

8. Answer A Concept 5

During an assessment, an analyst at a mid-size insurer notes that a public web server runs software with a known, unpatched flaw. ~~The server sits in a data center that was renovated in 2024.~~ In risk terms, what is the unpatched flaw?

Target word: classify the flaw **Decides:** A flaw is a weakness.

A	Correct	A weakness that could be exploited.
B	Adjacent category	Risk combines the likelihood and impact of a threat exploiting the weakness.
C	Confusion set	A threat is the actor or event that might exploit the weakness.
D	Absolute language	A flaw creates exposure; it does not prove compromise.

9. Answer B Concept 6

A security team at a logistics firm confirms that ransomware is encrypting files on two file servers. ~~The firm's cyber insurance renews next month, and the CEO is traveling.~~ Following the standard incident response lifecycle, what should the team do next?

Target word: next phase after confirmation **Decides:** The incident is confirmed and still spreading.

A	Inverted operation	Post-incident activity comes last, not while the attack is active.
B	Correct	Containment follows detection and analysis and stops further spread.
C	Dropped element	Skips containment and eradication; restored systems can be re-infected.
D	Camouflage	Coverage and travel do not change the next response step.

10. Answer C Concept 6 · transfer

An incident team has isolated an infected host and removed the malware and the account it used. Which phase comes next?

Target word: phase after eradication **Decides:** Containment and eradication are both done.

A	Dropped element	Skips recovery; the host is not back in service yet.
B	Confusion set	Preparation is the ongoing phase before incidents.
C	Correct	Recovery follows eradication.
D	Inverted operation	That phase came earlier.

11. Answer D Concept 7

A cloud hosting company is choosing a hypervisor for 300 production servers. ~~Its lead architect prefers the vendor with the best-rated support desk.~~ Security wants the smallest attack surface and no general-purpose host operating system under the virtual machines. Which choice fits?

Target word: hypervisor type **Decides:** No host operating system underneath.

A	Camouflage	Support quality does not meet the stated security requirement.
B	Absolute language	Containers share a kernel; they generally isolate less strongly than VMs.
C	Inverted operation	Type 2 runs on top of a host OS, adding attack surface.
D	Correct	It runs directly on hardware with no general-purpose OS below it.

12. Answer C Concept 8

A software team at a payments company runs a security scan only in the week before each release. Last quarter it shipped 14 releases, and late findings delayed three of them. Which change best lowers the cost of fixing security defects?

Target word: change that lowers fix cost **Decides:** Defects are found only at the end of the lifecycle.

A	Inverted operation	Removes the control instead of moving it earlier.
B	Adjacent category	Still a late gate; defects are found at the most expensive point.
C	Correct	Shifting security left finds defects when they are cheapest to fix.
D	Camouflage	Answers the release count, not when defects are found.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Defense in depth	Layer independent administrative, technical, and physical controls so that one failed control does not expose the asset.	One strong perimeter control (a top-rated firewall) is enough.	One control was bypassed and the question asks what would have limited the damage.
2	Zero trust	No request is trusted because of network location; every access is authenticated, authorized with least privilege, and continuously re-evaluated.	Anything inside the corporate network or VPN is trusted.	Access is granted because a user or device is “on the network.”
3	Protecting data by state	Data at rest needs storage encryption, data in transit needs channel encryption (TLS/IPsec), and data in use needs access and memory protections.	Disk encryption also protects data moving across the network.	Where the data is at the moment of exposure: stored, moving, or being processed.
4	Cloud shared responsibility	The provider secures the cloud; the customer secures what it puts in the cloud. The customer's share shrinks from IaaS to PaaS to SaaS, but data, identities, and access settings always stay with the customer.	Moving to the cloud hands all security to the provider.	“Who is responsible for patching / configuring ...?” plus a service model.
5	Threat, vulnerability, and risk	A vulnerability is a weakness; a threat is an actor or event that could exploit it; risk is the likelihood and impact of that happening.	Using threat, vulnerability, and risk as synonyms.	An unpatched flaw, misconfiguration, or weakness described in isolation.
6	Incident response lifecycle	Preparation → detection and analysis → containment, eradication, and recovery → post-incident activity; contain before you eradicate and recover.	Wiping and restoring immediately, or skipping straight to lessons learned.	“What should the team do next?” after an incident is confirmed.
7	Hypervisor types	A Type 1 (bare-metal) hypervisor runs directly on hardware with a small attack surface; a Type 2 (hosted) hypervisor runs on top of a general-purpose operating system.	Treating hosted desktop virtualization as the production-grade choice.	“No host operating system underneath” or “smallest attack surface.”
8	Secure development (shift left)	Build security into requirements, design, and every build; defects found earlier cost far less to fix than defects found at release.	Security testing is a final gate just before release.	Late findings delaying releases; “when in the lifecycle ...?”

Course scope (public description): Cybersecurity Architecture and Engineering provides learners with advanced skills and knowledge to design secure enterprise architecture solutions. The course focuses on assessing cybersecurity readiness and implementing enterprise- wide solutions to protect data and comply with an

organization's policies and frameworks. Course topics include integrating software applications, applying enterprise data security controls, evaluating cloud and virtualization solutions, analyzing threats and vulnerabilities, and responding to incidents.